Angelos Poulis

Boston, MA, USA

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Education

Boston University (BU) *Boston, MA*

Ph.D. in Computer Science *Sep 2024 – Present*

- *Advisors*: Evimaria Terzi, Mark Crovella
- Research interests: Interpretability of large language models
- Anticipated Graduation: May 2029

National and Kapodistrian University of Athens (NKUA) *Athens, Greece*

B.Sc. in Informatics and Telecommunications *Sep 2019 – Mar 2024*

- Major: Computer Science, Specialization: Data and Knowledge Management
- GPA: 8.7/10 (Honors)
- *Thesis*: "Reasoning over Description Logic-based Contexts with Transformers"
- *Supervisor*: Manolis Koubarakis, Co-supervisor: Eleni Tsalapati

Research Experience

Boston University *Boston, MA*

Graduate Research Assistant *Sep 2024 – Present*

- Studying the geometry of truth directions in large language models across factual recall and arithmetic reasoning tasks.
- Conducted probing analyses over model activations to evaluate generalization.
- Analyzed how task type, task difficulty, and model instructions affect truth direction generalization.
- *Advisors:* Evimaria Terzi and Mark Crovella.
- *Related output:* Manuscript under review, arXiv:2604.03754.

National and Kapodistrian University of Athens *Athens, Greece*

Research Intern, KDD lab *Oct 2023 – Jul 2024*

- Contributed to the AutoFair project, developing an automated toolkit for fairness evaluation of AI/ML algorithms through bias detection, fairness trade-off analysis, certification, and explainability.
- Helped develop an end-to-end pipeline for evaluating model fairness across data, model, and protected-attribute settings.
- *Supervisor:* Dimitrios Gunopulos.
- *Related output:* EuroSys Posters Track, 2024.

National and Kapodistrian University of Athens *Athens, Greece*

Undergraduate Research Student, AI Team *Jan 2022 – Feb 2024*

- Created DELTA$_D$, a 384K-example synthetic benchmark for evaluating natural-language reasoning over Description Logic-based contexts.
- Fine-tuned transformer language models (RoBERTa and DeBERTa), and evaluated GPT models in zero- and few-shot settings.
- Analyzed model generalization across reasoning depth, linguistic complexity, and symbolic variations.
- *Supervisor:* Manolis Koubarakis.
- *Related output:* EKAW 2024; earlier version presented at the NeLaMKRR Workshop at KR 2024.

Publications and Manuscripts

Under Review and Preprints

A. Poulis, M. Crovella, E. Terzi. "Testing the Limits of Truth Directions in LLMs." Manuscript under review. Preprint, arXiv:2604.03754, 2026.

Peer-Reviewed Publications

A. Poulis, E. Tsalapati, M. Koubarakis. "Transformers in the Service of Description Logic–Based Contexts." In *Proceedings of EKAW 2024*, LNAI/LNCS, Springer, 2025. Earlier version presented at the NeLaMKRR Workshop at KR 2024.

Research Presentations

A. Poulis, M. Crovella, E. Terzi. "Testing the Limits of Truth Directions in LLMs." *NEMI Workshop*, 2026.

A. Poulis, M. Crovella, E. Terzi. "Using Geometries of Truth to Identify Factual Recall in LLMs." *NEMI Workshop*, 2025.

A. Poulis, D. Tomaras, V. Kalogeraki, D. Gunopulos. "AutoFair Toolkit: An Automated Tool for Evaluating Fairness in AI/ML Models." *EuroSys Posters Track*, 2024.

Honors and Awards

2025 **Scholarship**, Gerondelis Foundation

2025 **Scholarship**, Eleni Gatzoyiannis Foundation

Spring 2024 **Dean's Fellowship**, Department of Computer Science, BU

Fall 2024 **Non-service Chair's Award**, Department of Computer Science, BU

2024 **Best Thesis Award**, Department of Informatics and Telecoms., NKUA

2018, 2019 **Cyprus Olympiad in Informatics Finalist**, Cyprus Computer Society

Teaching Experience

Spring 2026 **BU CS132: Geometric Algorithms (Linear Algebra)**, Graduate Teaching Fellow — *Boston, MA*

Fall 2023 **NKUA K22: Operating Systems**, Teaching Assistant — *Athens, Greece*

Outreach

2020–Present **Cyprus Olympiad in Informatics**, Scientific Committee Member

Nov 24–29, 2021 **Cyprus Olympiad in Informatics**, Team Deputy Leader

Skills / Technologies

Python • C/C++ • Java • SQL • PyTorch • scikit-learn • NumPy • Git